Chem Lab CSE KEY!

Summer 2025

Instructions:

- ➔ You will have 50 minutes to complete your exam. At the end of this time period, you must stop working. Partial credit will be awarded, so attempt as many questions as you can!
- ➔ [REFERENCE MATERIALS]Each team is allowed 2 note sheets with information on both sides, it must be printed and laminated or in a sheet protector.
- ➔ Ties will be broken in accordance with the national Science Olympiad rules in this order:
  - ⧫ Tiebreaker Questions answered Correctly

Name(s): _____

Score: __/60

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Questions? Comments? Email me:
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KEY:

1. c
2. b
3. c
4. b
5. d
6. b
7. c
8. d
9. c
10. d
11. a
12. c
13. b
14. d
15. c
16. b
17. b
18. a
19. a
20. d
21. b
22. a
23. c
24. c
25. b
26. d
27. d
28. a
29. d
30. d

FRQs:

1. Initial Rate Calculation

Rate law: rate = $k[A][B]^2$

Rate = $0.500 \times (0.2) \times (0.1)^2 = 0.500 \times 0.2 \times 0.01 = 0.0010$ mol·L^{-1}·s^{-1}

2. First-Order Rate Constant

Use:
$[A]=[A]_0e^{-kt}$

$\ln(0.4/0.8) = -kt$

$\ln(0.5) = -k(300)$

$k = 2.31 \times 10^{-3}\ s^{-1}$

3. Combustion of 2-Propanol

Balanced Equation:

$C_3H_8O + 4.5O_2 \rightarrow 3CO_2 + 4H_2O$

Net Ionic Equation: Same as molecular (combustion is molecular).

Molar Mass of 2-Propanol: ≈ 60.1 g/mol

Moles of 2-propanol = 100 g / 60.1 g/mol ≈ 1.664 mol

O_2 needed = 1.664 mol × 4.5 = 7.49 mol O_2

4. Rate Doubling/Quadrupling

Rate law: rate = $k[NO]^2[Cl_2]$

- Doubling $[Cl_2]$ doubles rate ⟹ first order in Cl_2
- Doubling [NO] quadruples rate ⟹ second order in NO

  Implication: Elementary step likely involves 2 NO molecules and 1 Cl_2 molecule interacting simultaneously, suggests a termolecular rate-determining step.

5. Molar Solubility in Common Ion (Na_2CrO_4)

$Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{2-}$

Let solubility of Ag_2CrO_4 = x

$[CrO_4^{2-}] \approx 0.2$ M (common ion)

$K_{sp} = [Ag^+]^2[CrO_4^{2-}] = (2x)^2(0.2) = 4x^2(0.2) = 0.8x^2$

Set $K_{sp} = 1.2 \times 10^{-12} = 0.8x^2$

$x^2 = 1.5 \times 10^{-12} \Longrightarrow x = \sqrt{(1.5 \times 10^{-12})} \approx 1.22 \times 10^{-6}$ M

6. Rate Constant Ratio (Arrhenius Equation)

$k_2/k_1 = e^{Ea/r(1/T1 - 1/T2)}$

Ea = 75,000 J/mol, R = 8.314 J/mol·K

$k_2/k_1 = e^{75000/8.314(1/300 - 1/320)}$

$k_2/k_1 = e^{1.878} =$ **6.54**

7. Limiting Reagent and Theoretical Yield

Reaction: $C_6H_5CH_2Cl + NaOH \rightarrow C_6H_5CH_2OH + NaCl$

Limiting reagent = Benzyl chloride

Product (1:1) ⟹ 0.0948 mol benzyl alcohol formed

Molar mass benzyl alcohol ≈ 108.1 g/mol

0.0948 mol × 108.1 g/mol ≈ 10.24 g